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Efficient Thermal/Energy Management in EVs at Low Temperatures

School of Aerospace, Transport and Manufacturing
Automotive Engineering

MSc
Academic Year: 2023 - 2024

Supervisor: Dr Abbas Fotouhi

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This thesis is submitted in partial fulfilment of the requirements for
the degree of MSc

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Academic integrity declaration

I declare that:

- The thesis submitted has been written by me alone.
- The thesis submitted has not been previously submitted to this university or any other institution.
- That all content, including primary and/or secondary data, is true to the best of my knowledge.
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Abstract

This thesis delves into the study of an Electric Vehicle Thermal Management System (EVTMS) with a focus on enhancing energy efficiency and thermal control in electric vehicles. The research aims to improve thermal management strategies for essential components such as the cabin, battery, and power electronics to ensure optimal performance under different environmental conditions. Detailed simulation work is done in Matlab/Simulink and by using the blocks from the Simscape library to study the basics of a refrigeration loop and implementation of components like condenser, evaporator, compressor, etc. The model is employed to simulate the EVTMS's thermal behaviour and energy consumption, considering various operating scenarios, including extreme temperatures. The findings highlight the significant influence of component-level study of heat flow rates and power consumption on the overall system's efficiency. The analysis specifically focuses on the heat flow rates of the condenser, evaporator, chiller, and coolant radiator, shedding light on their roles in maintaining target temperatures within the cabin and battery compartments. The obtained heat flow rates indicate the fluctuations in the system, which shows the heat loss. The power consumption profiles of the compressor, condenser fan, blower fan, coolant pump, and radiator fan were assessed, providing insights into crucial energy usage points that impact the system's performance. Additionally, the model is just a test bench for studying an electric vehicle thermal management system. You can also apply real-life scenarios to the model to study the cabin load and comfort.

Keywords:

Matlab, Simulink, Positive Temperature Coefficient, Thermodynamics, Electric Vehicle

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List of Abbreviations

IT	Information Technology
EVTMS	Electric Vehicle Thermal Management System
TMS	Thermal Management System
EV	Electric Vehicle
ICEV	Internal Combustion Engine Vehicle
PTC	Positive Temperature Coefficient
TXV	Thermal Expansion Valve
COP	Coefficient of Performance
HVAC	Heating, Ventilation and Air Conditioning
HEX	Heat Exchanger
PID	Proportional Integral Derivative

1 Chapter 1:

Introduction

1.1 General Background

As the world moves toward more environmentally friendly forms of transportation, electric cars (EVs) have become essential. They offer significant reductions in greenhouse gas emissions compared to conventional internal combustion engine vehicles (ICEVs). However, EVs face unique challenges, particularly in thermal management, which significantly impact vehicle efficiency, performance, and passenger comfort. Efficient thermal management systems are critical for optimising the energy consumption of EVs, especially under extreme temperature conditions where heating and cooling demands are substantial.

Electric vehicles have a greater potential for relieving environmental problems. The air conditioning system has an important role in an EV, which keeps drawing research concerns about its energy consumption and refrigerant issues [1]. Since 2021, first-quarter electric car sales have typically accounted for 15-20% of global annual sales. Based on this observed trend, policy momentum and the seasonality that EV sales normally experience, we estimate that electric car sales could reach around 17 million in 2024. This indicates robust growth for a maturing market and underscores the immense potential of the electric car market[2].

Electric car sales, 2010-2024

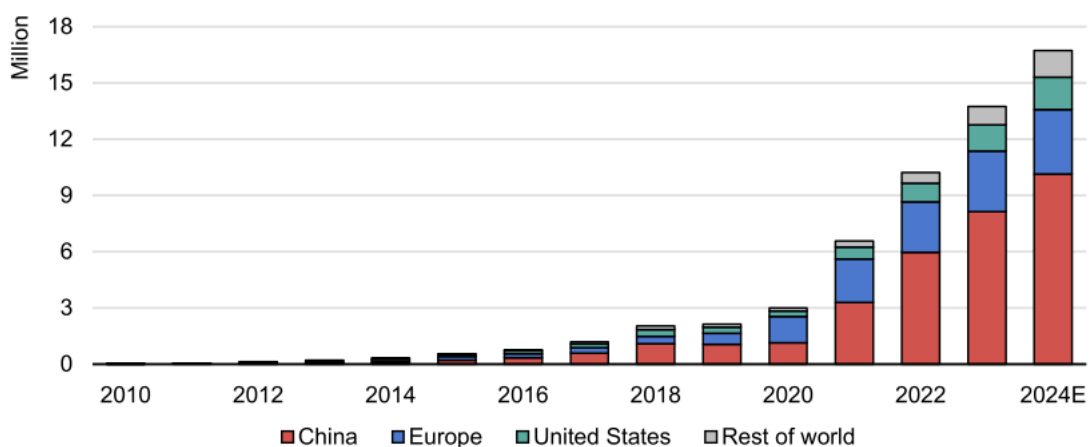


Figure 1: EV sales

Error! Reference source not found. shows the rise of Electric car sales from 2010 to the present. The sales are estimated based on the different trends in the market, and recent trends and impending CO2 targets for 2025 suggest that Europe will see the slowest growth in electric car sales among the three largest markets, with an estimated 3.5 million units sold in 2024, representing less than 10% growth from the previous year. Despite a generally weak passenger car market, electric vehicles will account for about one in four cars sold in Europe. Conversely, outside these significant markets, electric car sales are projected to exceed 1 million units in 2024, a notable increase of over 40% from 2023. The success of domestic and Chinese electric car manufacturers in Southeast Asia underscores the region's significant contribution to emerging EV markets. Despite uncertainties regarding India's FAME III scheme and potential subsidies, electric car sales in India are expected to remain strong, growing by around 50% compared to 2023. Overall, in regions outside the top three EV markets, electric vehicles are anticipated to represent about 5% of total car sales in 2024, suggesting a nearing global tipping point toward mass EV adoption[2].

The thermal management system in an EV in cold conditions mainly comprises battery, powertrain, and cabin heating. Each system has different thermal demands, which are also transient in behaviour. Controlling these various levels of thermal loads at any point in an EV application is not easy. The different demands of each thermal system make the overall thermal management system more complex during operation. Heat Pump systems have shown promise in extending the driving range of EVs by utilising ambient heat for cabin heating, thereby reducing the energy drain on the battery. For instance, studies have demonstrated that heat pump systems can increase driving range by up to 23% compared to traditional positive-temperature-coefficient (PTC) heaters at -10°C [3][4]. These systems have the ability to offer heating and cooling, guaranteeing passenger comfort in different environmental situations. Nonetheless, heat pump systems are receiving increased favourable recommendations for their heating abilities in cold conditions.

The ongoing research and advancements in EV thermal management systems are crucial for addressing electric mobility's efficiency and performance challenges. By leveraging innovative technologies and sophisticated modelling techniques, thermal management solutions can be developed that enhance vehicle range and passenger comfort and contribute to energy conservation and emission reduction goals[5], [6].

This paper discusses the implementation of the refrigeration cycle in a thermal management system, which functions to heat the cabin and maintain the cabin temperature. A PTC heater is used for this operation, and the heat flow rates of the system components are obtained, which helps in studying the energy losses in components.

1.2 Aim and objectives

This thesis aims to explore EV thermal management strategies and design a model-based control strategy for efficient energy management. The major objectives of this thesis are:

1. To gain knowledge of an EV thermal management system from the perspective of thermodynamics and thermo-economics.
2. To review the implementation of an Electric PTC heater, Heat pump, Heat battery, and other waste heat recovery techniques for EV application.

2 Chapter 2:

Literature Review

2.1 Thermal Management in EV

The thermal management system is one of the most important parts of an electric vehicle, and it majorly compromises battery cooling and powertrain cooling.[7] .The production of CO₂ has always been a major talking point in terms of environmental impacts, but electric vehicles are deemed to be the vehicles that can reduce CO₂ notably [4]. Due to the need for better output performance for these thermal management systems with less available space in the vehicle, heat dissipation performance has become crucial[8]. Apart from the cooling, the EVTMS, in cold conditions, faces many significant problems, such as cabin heating, battery heating, and range anxiety. One of the main challenges is that the batteries at low temperatures tend to have a lower life span. This may occur due to many factors, such as temperatures, discharge rate, and depth of discharge [9]. The change in ambient temperature affects the systems, mainly the battery life. Heating the battery so that it operates in optimal conditions is needed for these systems to function in a stable state. In the Northern parts of the world, where the temperature during winter is between -20 deg /- 10 deg, driving an electric car has its difficulties[9]. The lithium batteries in electric vehicles have come to a lot of research for their high energy density and power [10] [11]. Unlike combustion engines, EVs must also look for cabin comfort at low temperatures[12]. The effects of low temperatures tend to impact the battery's available energy and also experience serious degradation[13].

2.2 Energy Management in EVs

Energy management in electric vehicles (EVs) is a critical factor in ensuring these vehicles' efficiency, performance, and sustainability. As EVs continue to gain popularity, the challenge of optimising energy use becomes increasingly important. This involves not only maximising the range of the vehicle but also ensuring passenger comfort and extending the lifespan of key components like the battery and power electronics. Effective energy management in EVs

integrates several systems, including thermal management systems (TMS), optimising heating, ventilation, and air conditioning (HVAC) systems, and innovative strategies to handle energy demands in various environmental conditions. The battery, motor, and power electronics in EVs generate a significant amount of heat during operation, and this heat must be managed effectively to maintain efficiency and prevent damage[14]. Thermal management systems are designed to keep these components within their optimal temperature ranges, which is essential for ensuring the vehicle's performance and safety. According to [8], various strategies such as liquid cooling, air cooling, and the use of heat exchangers are employed to manage heat effectively. A study assessing these thermal management systems further explores their effectiveness [15]. This study compares technologies that cool and heat the battery, motor, and power electronics, demonstrating how an effective TMS can significantly impact overall energy consumption and vehicle performance. Poor thermal management can lead to increased energy consumption, reduced battery life, and compromised vehicle safety, making EV design a critical area of focus.

2.3 HVAC system and Vehicle Cabin load

The heating, ventilation and air conditioning (HVAC) system has an impact on the energy management of the vehicle. Mainly, when you consider heating in these vehicles, the HVAC system is much more different from ICEV. Thermal loads of a vehicle cabin have their effects on the power consumption of the HVAC system and the cabin load of an EV, and HEV is more similar as far as the cabin does not differ from each other[16].

Under varied vehicle operating conditions, an automotive HVAC system's primary goal is to provide thermal comfort to its occupants. With transient environmental conditions, air temperature is one of the main factors that significantly affects thermal comfort[17].

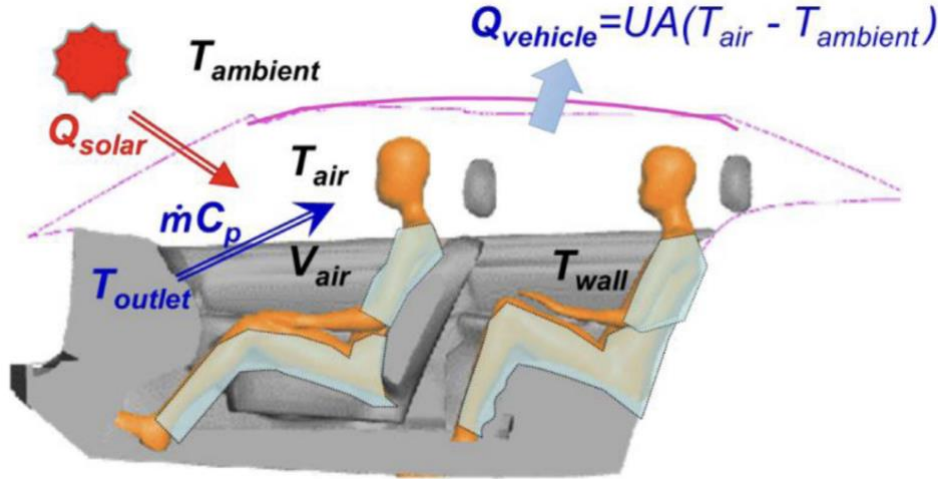


Figure 2: Cabin thermal load

2.4 Challenges at Low Temperatures

One of the critical issues EV owners face in low-temperature environments is range anxiety and energy consumption. One of the main reasons an EV consumes a lot of power is for cabin comfort, and for this function, it uses either a PTC heater or a heat pump. The usage of a PTC heater as a primary component for cabin heating is common as it is low in cost and easily controlled compared to a heat pump system[18][10]. The PTC heater's function is to convert battery electricity into thermal load, which consumes more energy from the battery. This impacts the vehicle's mileage, acceleration, and battery efficiency. On the other hand, a heat pump system utilises the refrigerant latent heat for heating, leading to high energy efficiency and reduced electricity consumption from the battery. However, it also has drawbacks. In severe cold weather, evaporator frost occurs, and compressor performance declines, leading to decreased heating system performance[6][19].

2.5 Use of PTC Heaters vs. Heat Pumps in Electric Vehicles

The effective regulation of temperature in electric vehicles (EVs) plays a crucial role in maintaining peak performance, efficiency, and passenger comfort, especially in diverse weather conditions. Two key technologies utilized for heating in EVs are Positive Temperature Coefficient (PTC) heaters and heat pumps. Each technology comes with its own set of advantages and limitations, impacting the overall energy efficiency and driving range of EVs.

2.5.1 PTC Heaters

PTC heaters are commonly employed in EVs due to their simplicity and dependable operation. These heaters operate by converting electrical energy directly into heat via resistive elements that increase in resistance as they generate heat. This self-regulating feature helps prevent overheating and ensures consistent performance. Nonetheless, despite their straightforward design, PTC heaters are known for their relatively high energy consumption, directly influencing the driving range of EVs. Research indicates that the energy efficiency of PTC heaters is restricted, especially in colder climates. For instance, a study comparing different heating methods in EVs revealed that the use of PTC heaters could result in a substantial decrease in driving range, particularly in sub-zero temperatures [4]. At -10°C , the driving range of an EV equipped with a PTC heater was found to be up to 39.9% lower than that of an EV equipped with a more efficient heating solution[3]. This is primarily due to PTC heaters relying solely on the vehicle's battery for energy without utilising external heat sources.

2.5.2 Heat Pumps

The present heat pump systems are more energy-efficient options for heating in EVs. Operating on the principle of transferring heat from the external environment into the vehicle, heat pumps are capable of effectively heating the vehicle even in relatively cold conditions. This approach is more efficient than direct electrical heating as it involves the transfer of heat rather than its generation, often achieving a coefficient of performance (COP) greater than 1, indicating that more heat is produced than the electrical energy consumed. The benefits of heat

pumps are particularly evident in cold climates, where they can significantly reduce the energy drain on the battery compared to PTC heaters. Studies have demonstrated that heat pumps can extend the driving range of EVs by up to 23% in comparison to vehicles using PTC heaters in similar ambient temperatures [3], [20]. This improvement is achieved by harnessing ambient heat, even in low temperatures, to warm the passenger cabin, thereby decreasing reliance on the battery. Furthermore, advanced heat pump systems in EVs are engineered to operate effectively in a broad range of conditions, providing both heating and cooling as needed[21], [22]. These systems often integrate additional components, such as internal heat exchangers and phase change materials (PCMs), to further enhance performance and efficiency [3]. For example, a well-designed heat pump system can fulfil the thermal comfort requirements of the passenger compartment while maintaining low energy consumption, a critical factor in extending the driving range[20].

3 Chapter 3:

Methodology

3.1 System Design

The model is designed with the purpose study the thermal/thermodynamics side of an EV thermal management system. It consists of a refrigeration loop and coolant loop. The refrigeration loop is a vapour compression cycle that uses the refrigerant(R-1234yf) to heat the cabin directly and indirectly heat the battery via the coolant circuit. The model uses an FTP75 drive cycle with reference speed and current as input.

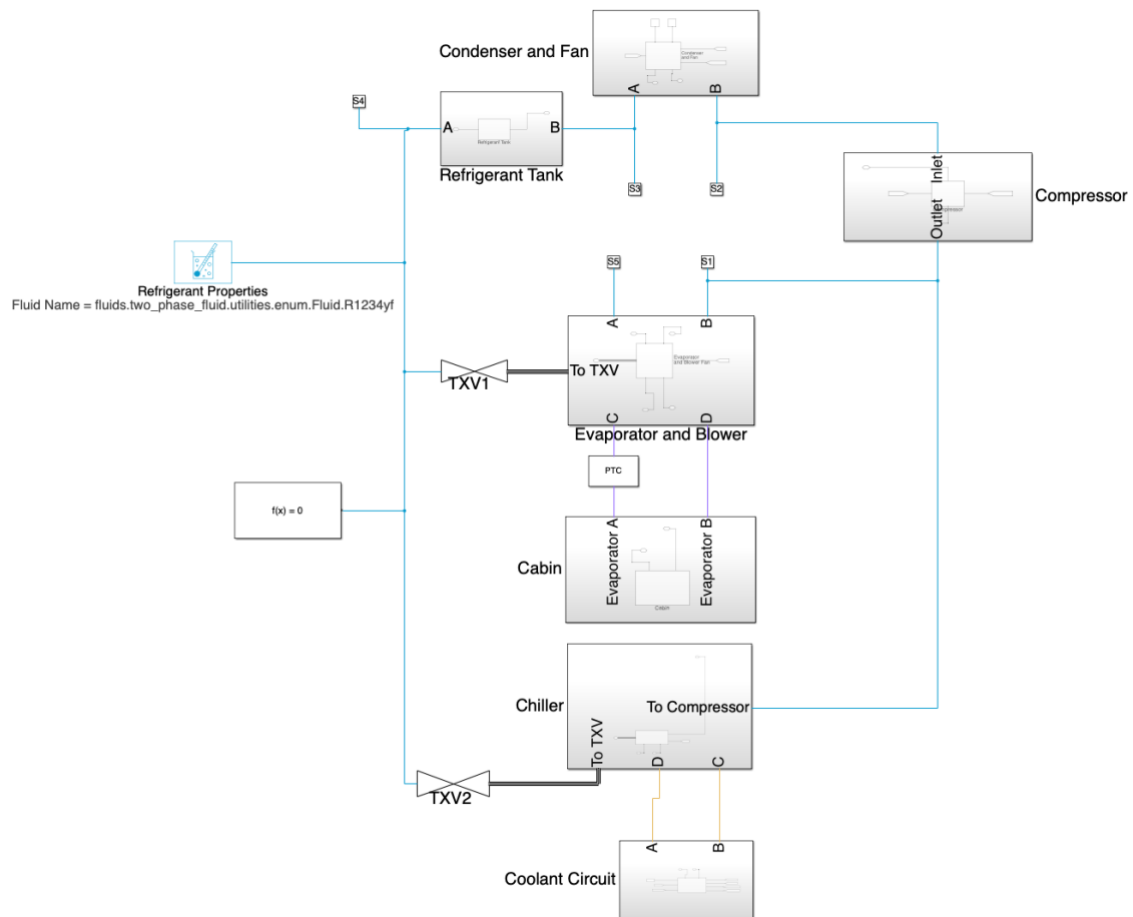


Figure 3: EVTMS model layout

Figure 3 shows the design layout of the EVTMS, which consists of seven subsystems. This system utilises a refrigerant-based cycle, ensuring the vehicle operates efficiently in different environmental conditions.

3.1.1 System Sizing

To calculate the heat load required for heating a volume of humid air, we will apply thermodynamic principles such as enthalpy and latent heat. And Use the Magnus equation to calculate the specific humidity at initial and final conditions[23].

- Consider sensible and latent heat when determining the enthalpy at initial and final conditions.
- Calculate the change in enthalpy.
- Estimate the total heat load needed to heat the volume.

Parameters required for the load calculation:

Table 1: Load calculation parameters

Parameters	Values
Initial temperature	-10 deg C
Final temperature	20 deg C
Relative Humidity Initial	90 %
Relative humidity Final	50 %
Volume	4 m ³
Time duration	60s
Pressure	101325 Pa

These parameters are assumed for calculation purposes.

The calculation for the system heat load is done using MATLAB script, and the script is added to the appendix.

- Specific humidity:

$$\omega = \frac{0.622 * RH * e_{sat}(T)}{P - (1 - RH) * e_{sat}(T)}$$

- Enthalpy:

$$h = C_p * (T + 273) + \omega * h_{fg}$$

- Change in enthalpy:

$$\Delta h = h_{final} - h_{initial}$$

- Total heat load:

$$\text{Heat load} = \Delta h * m_{air}$$

- Heating power:

$$\text{Heating power} = \frac{\text{Heatload}}{\text{Time}}$$

The time taken is considered to be 1000s as it is the time duration of the FTP75 drive cycle.

The result for the heat calculation:

The Total heating load = 162.58 kJ

Total heating power = 2.71 kW

The system is sized as it requires a total power of 2.71 kW.

3.2 Coolant circuit

This subsystem is engineered to regulate the temperature of vital vehicle components, such as the battery, motor, inverter, and associated electronics, ensuring their optimal performance and longevity.

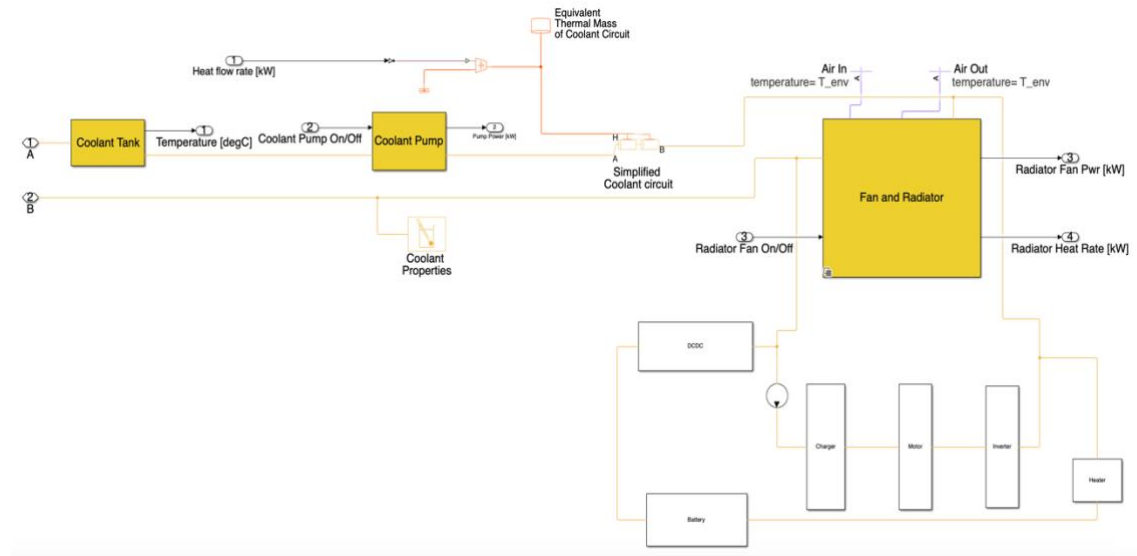


Figure 4: Modelling of the coolant circuit

The EVTMS's coolant circuit is a complex, integrated system that is intended to manage the thermal loads of an electric vehicle's essential components. The system keeps the best possible operating conditions by making sure that heat is transferred from the battery, motor, inverter, and other electronics to the outside environment efficiently. The longevity and general performance of the vehicle are enhanced by responsive fan operation and controlled coolant flow, which combine energy efficiency with excellent cooling/heating.

3.2.1 Coolant System Sizing

For heating the coolant, we need to size the power required by the coolant circuit. The coolant is ethylene glycol and water:

- Change in Temperature:

$$\Delta T = T_{final} - T_{initial}$$

- Coolant mixture specific heat capacity:

$$C_{p,mix} = W_{eg} * C_{p,eg} + W_{wat} * C_{p,wat}$$

- Heat load:

$$\text{Heat load} = m_{coolant} * C_{p,mix} * \Delta T$$

- Heating Power:

$$\text{Heating power} = \frac{\text{Heatload}}{\text{Time}}$$

The time taken is considered 1000s as it is the duration of the FTP75 drive cycle.

The result for the coolant system sizing:

The Total heating load = 495.45 kJ

Total heating power = 0.50 kW

The system is sized as it requires a total power of 0.50 kW

3.2.2 Coolant Pump

The primary function of the Coolant Pump is to circulate fluid within the EVTMS. It adjusts its operation based on a duty cycle controlled by the thermal management controller, which receives input from temperature sensors located at key points in the circuit. The power consumption of the pump, usually measured in kilowatts (kW), depends on both the flow rate of the coolant and the pressure difference it creates within the system. This pump is responsible for propelling the coolant through the entire circuit, maintaining the required flow rate to ensure effective heat transfer from high-temperature components.

When in heating mode, the pump guarantees that hot coolant reaches all the components that need heating, like the battery, motor, or cabin heater. In cold environments, the pump may need to operate more frequently or continuously to maintain the required temperatures.

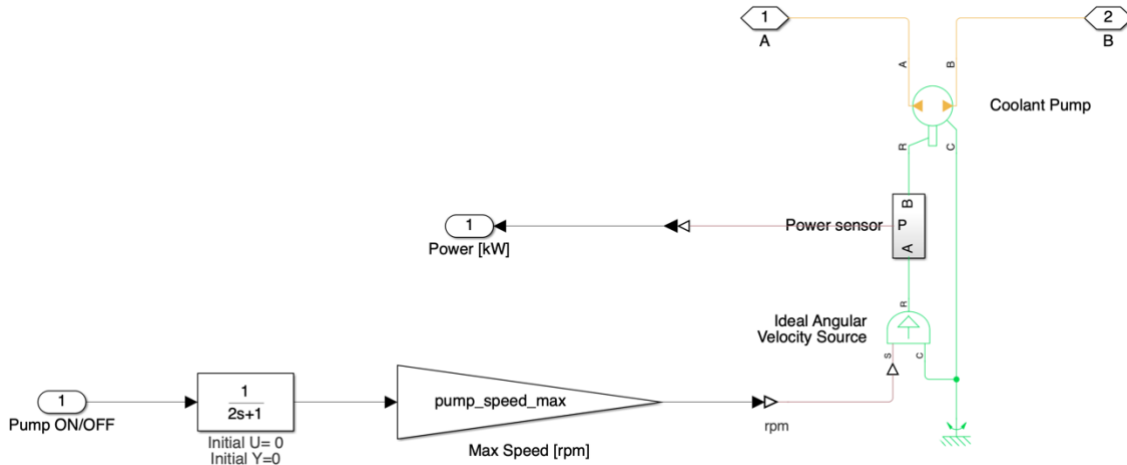


Figure 5: Coolant pump model

3.2.3 Coolant Tank

The coolant tank serves as a storage unit for coolant, which is typically a mixture of water and glycol. It plays a crucial role in maintaining a steady supply of coolant for the cooling system. The tank is designed with multiple inlet and outlet ports (labelled A1, A2, B1, and B2), which enable the integration of different coolant flows and ensure uniform management of coolant level and temperature. Additionally, the tank has a temperature sensor that provides real-time monitoring of the coolant's temperature. This feature is essential for ensuring that the system's thermal conditions remain optimal. Implementing the coolant tank for the system helps to supply the circuit components with a stable supply of coolant. As the coolant circulates through the system, it absorbs heat generated by these components, preventing overheating and maintaining efficient operation. The temperature data from the sensor can be used to adjust the operation of pumps

and fans in the control system, ensuring that the coolant stays within the desired temperature range.

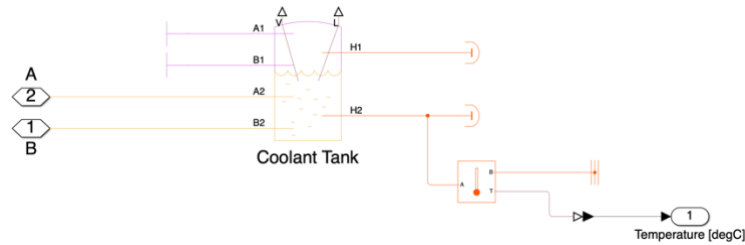


Figure 6: Coolant Tank model

3.2.4 Radiator and Fan

The Radiator and Fan subsystem plays a crucial role in the EVTMS by helping to dissipate heat. When the coolant, which now contains thermal energy, enters the radiator, heat exchange occurs through convective and conductive processes. This is made possible by the surface area of the radiator fins and the airflow generated by the radiator fan. The fan is turned on when a certain thermal threshold is reached, and its power usage is adjusted to strike a balance between energy efficiency and cooling needs. The effectiveness of the radiator is determined by its thermal resistance and the temperature difference between the incoming and outgoing air. The heat is dispersed into the surrounding air by the radiator. This is done by using the fan to push air through the radiator. This effectiveness is often quantified regarding the radiator's heat rate, measured in kW[24].

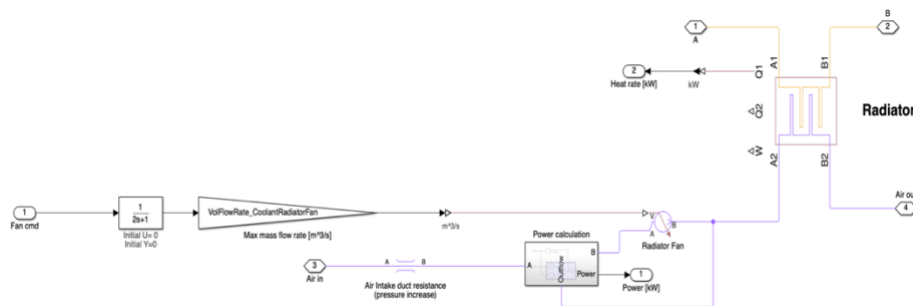


Figure 7: Radiator and Fan model

3.2.5 Integration with vehicle components

The coolant system is closely connected to the main power and energy storage parts of the vehicle, including the Battery, Motor, Inverter, DC-DC Converter (DCDC), and Charger. This system is designed to absorb excess heat produced during high-demand situations like quick acceleration or fast charging. Additionally, the inclusion of a Heater in the system helps manage temperature during cold conditions, making sure that the battery and power electronics stay within the best temperature ranges for operation[25].

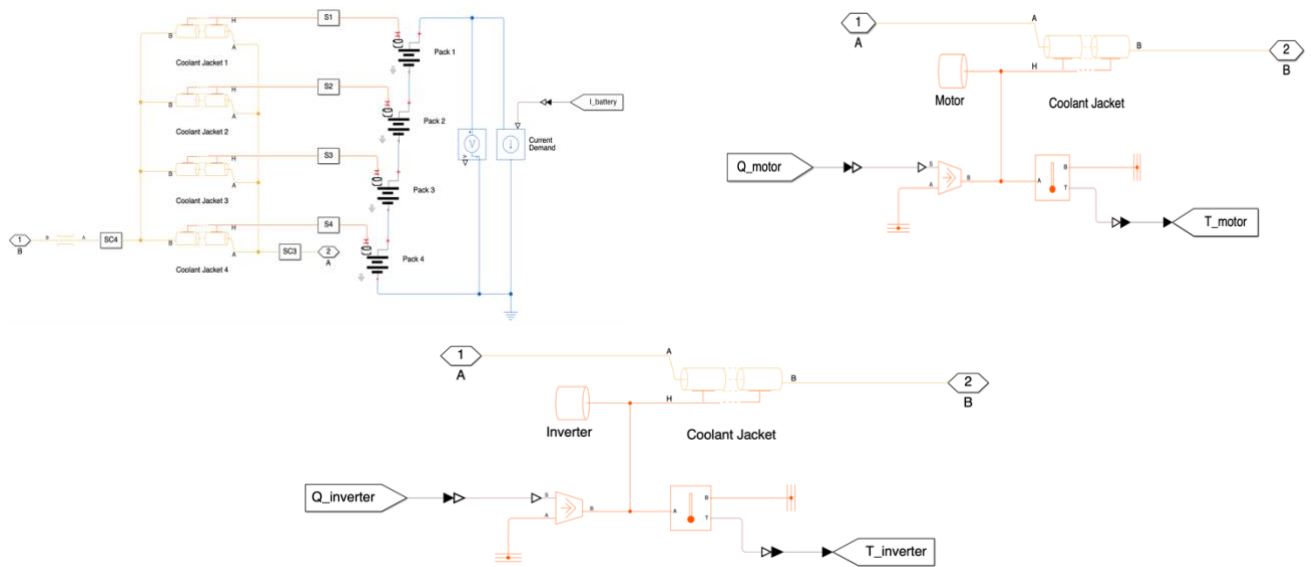


Figure 8: Battery, motor, inverter model in Simulink/Simscape

Figure 8 depicts the system-level blocks for the battery, motor, and inverter, which are the primary sources of heat in an EV. The battery under consideration consists of four modules, each comprising 20 cells. While a detailed battery model is possible, this thesis utilises a simplified level battery model for its analysis.

3.3 Refrigeration loop

The refrigeration loop is modelled based on the refrigerant's pressure and enthalpy diagram (R-1234yf). This determines the cycle, and each component function is specified[1].

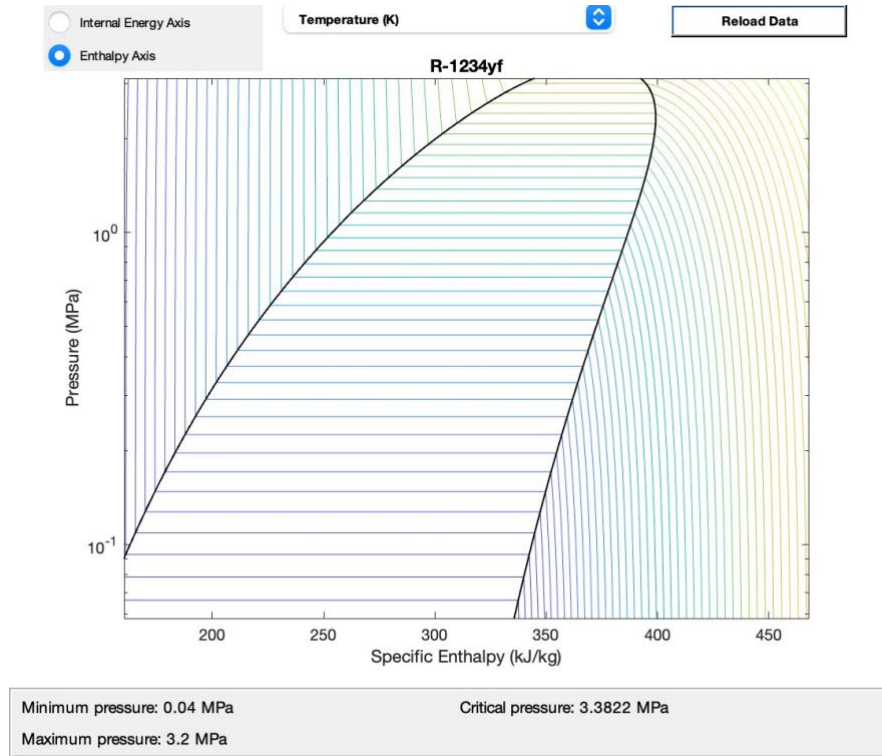


Figure 9: Pressure vs Enthalpy graph of R-1234yf

3.3.1 Compressor

The compressor system plays a vital role in the refrigeration cycle for heating and cooling; it drives the refrigeration cycle. It takes in low-pressure refrigerant vapour and compresses it to a high-pressure and high-temperature state, which is essential for effective heat transfer. Depending on the system's mode (heating or cooling), this process allows the refrigerant to release or absorb heat, ensuring that the vehicle's cabin and battery are maintained at optimal temperatures.

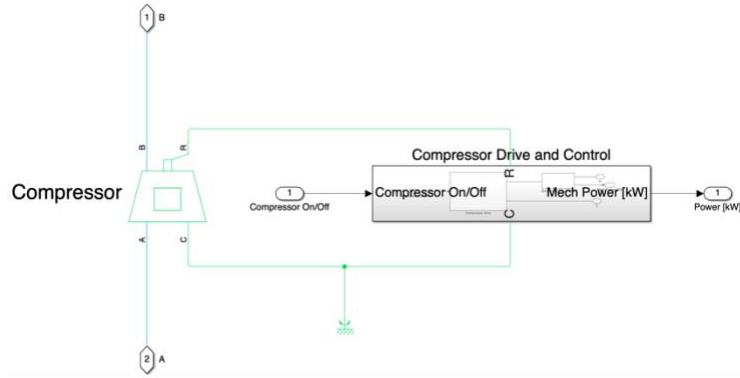


Figure 10: Compressor model

The compressor drive and control system shown in the diagram is designed to regulate the rotational speed of the compressor's shaft, which is crucial for controlling the compressor's operation within the refrigeration loop. The system starts with a signal that determines whether the compressor should be active or not. This signal goes through a transfer function block ($1/2s+1$), representing a first-order lag that simulates the compressor's response time to changes in the input. This block's output is the compressor shaft's nominal rotational speed (RPM_Compressor_nominal). A saturation block is utilised to maintain numerical stability and prevent the system from reaching unrealistic conditions, which sets lower and upper limits on the RPM. The lower limit prevents the RPM from dropping below a minimum threshold. The controlled RPM is then supplied to the compressor as a speed source, ensuring that the compressor operates within its optimal speed range to balance the system's thermal load demands and mechanical efficiency.

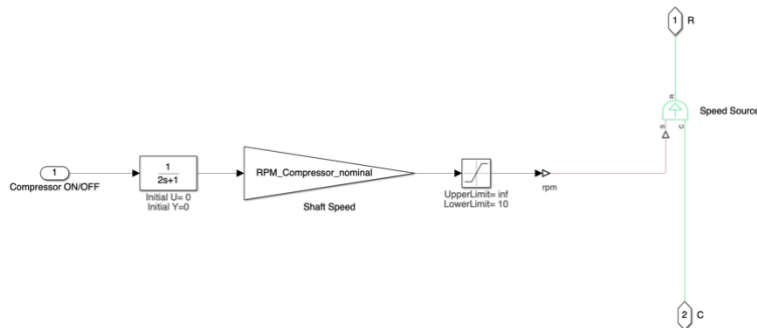


Figure 11: Compressor drive block

3.3.2 Evaporator and Blower

The evaporator is used in the system for facilitating heat exchange between the refrigerant and the air flowing from the cabin. The system uses the blower fan to pull warm air from the cabin and blow it over the evaporator coils. The refrigerant, flowing through the evaporator, absorbs the heat from the air, cooling it in the process. The cooler air is then returned to the cabin to maintain the desired temperature. The Blower Fan, On/Off control, determines the operation of the fan, which is modulated by a first-order transfer function to manage the airflow rate smoothly, ensuring stable operation. The fan's performance is influenced by the air intake duct resistance, which accounts for pressure drops, and the fan's power consumption is calculated and monitored in real time to optimise energy efficiency. Additionally, the evaporator is connected to the Thermostatic Expansion Valve (TXV), which regulates refrigerant flow into the evaporator, ensuring efficient heat absorption while maintaining system stability.

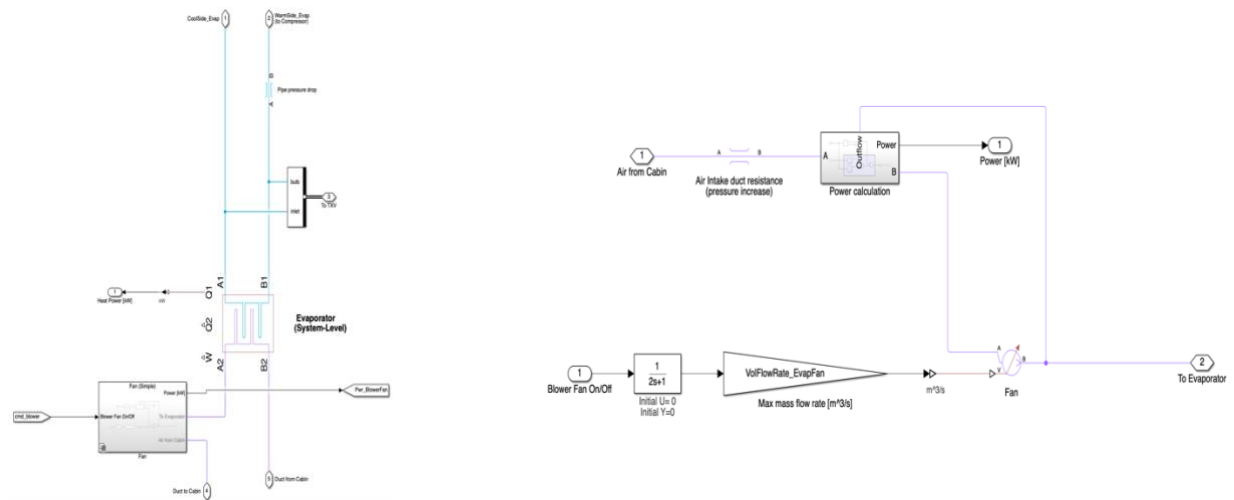


Figure 12: Evaporator and Blower

3.3.3 Thermostatic Expansion Valve

The TXV's bulb has a temperature-sensitive material, like refrigerant, which expands, or contracts based on the temperature. This controls the opening and closing of the TXV valve, regulating the refrigerant flow into the evaporator coil.

The bulb temperature is usually measured at the evaporator coil outlet. The bulb is attached to the suction line at the outlet of the evaporator, allowing it to sense the refrigerant vapour temperature leaving the evaporator. By measuring the temperature at the evaporator outlet, the TXV can maintain a consistent superheat level, ensuring optimal evaporator performance and preventing issues such as flooding or starving the evaporator coil. The superheat level is the difference between the measured bulb temperature and the refrigerant saturation temperature at the evaporator outlet.

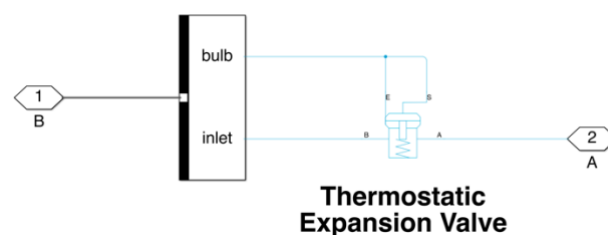


Figure 13: Thermal Expansion Valve

3.3.4 Condenser and Fan

The condenser and fan model play a crucial role in releasing the heat absorbed by the refrigerant during the compression cycle. The condenser receives high-pressure, high-temperature refrigerant vapor from the compressor, and the condenser fan helps facilitate the heat exchange process. The fan draws air through the system, increasing airflow over the condenser coils, which aids in cooling and condensing the refrigerant into a liquid state. Additionally, there is a Fan On/Off control to regulate the operation of the condenser fan and ensure optimal performance. This integration within the refrigeration loop is essential for

maintaining the cycle's efficiency and supporting the overall thermal management system's effectiveness in maintaining desired temperatures within the vehicle.

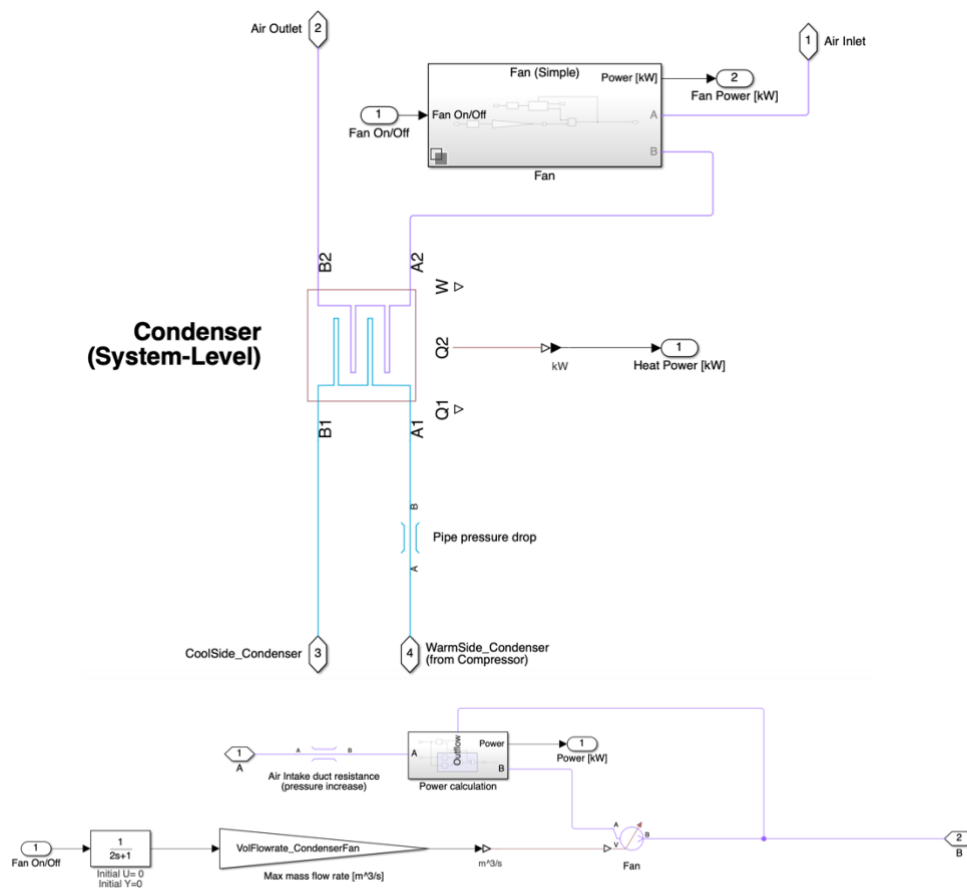


Figure 14: Condenser and Fan model

3.3.5 Chiller

The heat exchanger allows the low temperature refrigerant to transfer the heat into the coolant which flows through the coolant circuit warming the coolant(water-glycol). This coolant flows through the circuit heating up the battery and other EV components. Just like the evaporator the refrigerant flow is controlled by the TXV.

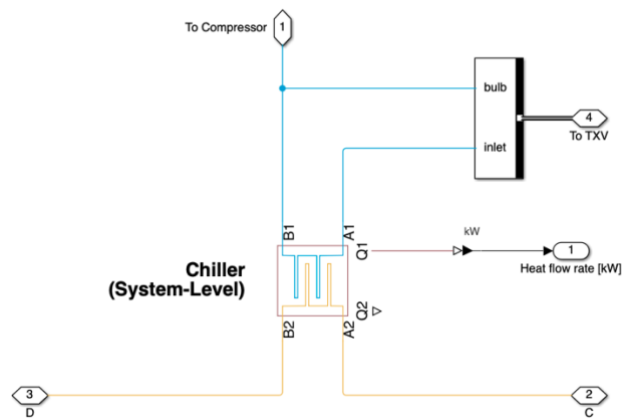


Figure 15: Chiller HEX model

3.4 Cabin Modelling

The cabin model system provides a detailed simulation of the thermal and environmental dynamics within a vehicle cabin. It considers external factors such as temperature, pressure, humidity, and CO₂ levels to define the characteristics of the air entering the cabin. The model also considers potential modifications due to leakage and the impact of occupant effects, including moisture, CO₂, and heat contributions. Additionally, the system simulates the thermal impact of solar radiation entering the cabin, further influencing the internal heat balance. Overall, it processes these inputs to adjust the cabin air's temperature and quality accordingly.

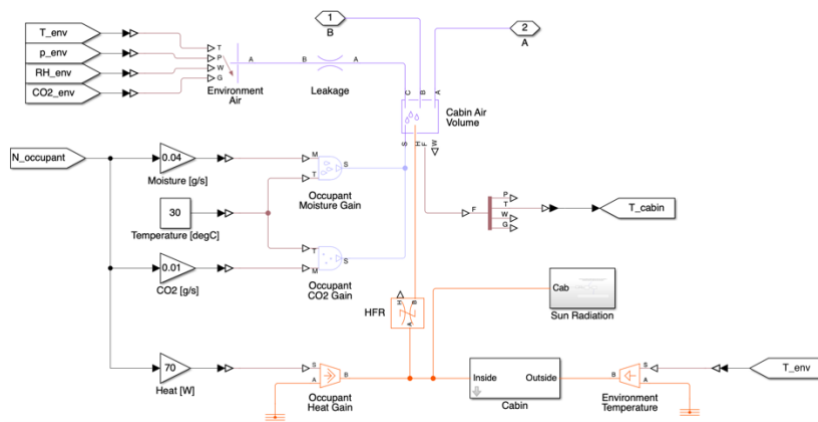


Figure 16: Cabin system model

The cabin's air temperature is adjusted to match the outside environment, considering heat flow and air circulation. This is essential for accurately replicating the thermal response of the cabin in various conditions, aiding in the understanding and enhancement of the vehicle's HVAC system. It ensures occupants are comfortable while maximising energy efficiency in various environmental situations.

3.5 Positive Temperature Coefficient Heater

For the objective of this thesis, a PTC heater is used to control and heat cabins. The activation of the PTC heater is controlled by the (cmd_ptc) signal, which modulates its operation through a first-order transfer function, resulting in a smooth response. The heater adjusts its power output up to a maximum of 2500 W, based on the system's demands through a Controlled Heat Flow Rate Source. As the heater generates heat, this energy is transferred into the PTC Air Volume block, which represents the surrounding air space. The heated air is then mixed and circulated throughout the cabin, interacting with the overall cabin model. The Pressure and Temperature sensor monitors the PTC heater's temperature, ensuring it operates within safe limits and provides feedback to adjust the heating power if necessary. In the cabin model, the PTC heater complements the main HVAC system by providing targeted heat when the cabin temperature (T_{cabin}) drops below a certain threshold, thereby enhancing passenger comfort. The interaction between the PTC heater and the cabin model creates a balanced and responsive climate control system capable of adapting to varying thermal loads and maintaining the desired cabin environment efficiently.

The control logic is designed to maintain the cabin temperature at a desired level, ensuring passenger comfort in different environmental conditions. To calculate the error, the system compares the current cabin temperature (T_{cabin}) with the setpoint temperature (T_{setpoint}). This error is then adjusted based on the difference between the environmental temperature (T_{env}) and the setpoint to account for external temperature influences. The resulting error signal is then processed by a PI (Proportional-Integral) controller, which then processes the resulting error signal after being fed into a multiplier block, which may adjust the signal based on additional factors. The PI controller refines the control signal to generate the blower command (cmd_blower), which adjusts the airflow into the cabin to either increase or decrease the temperature.

At the same time, the system also monitors the PTC heater temperature (T_{ptc}). If the temperature is below a certain threshold (indicated by the zero reference), a comparator activates the PTC heater command (cmd_ptc) to provide supplemental heating. This dual control strategy allows for precise regulation of the cabin environment by dynamically adjusting both the blower speed and the activation of the PTC heater based on real-time conditions. This helps ensure that the cabin temperature remains stable and within the desired range despite fluctuations in external conditions.

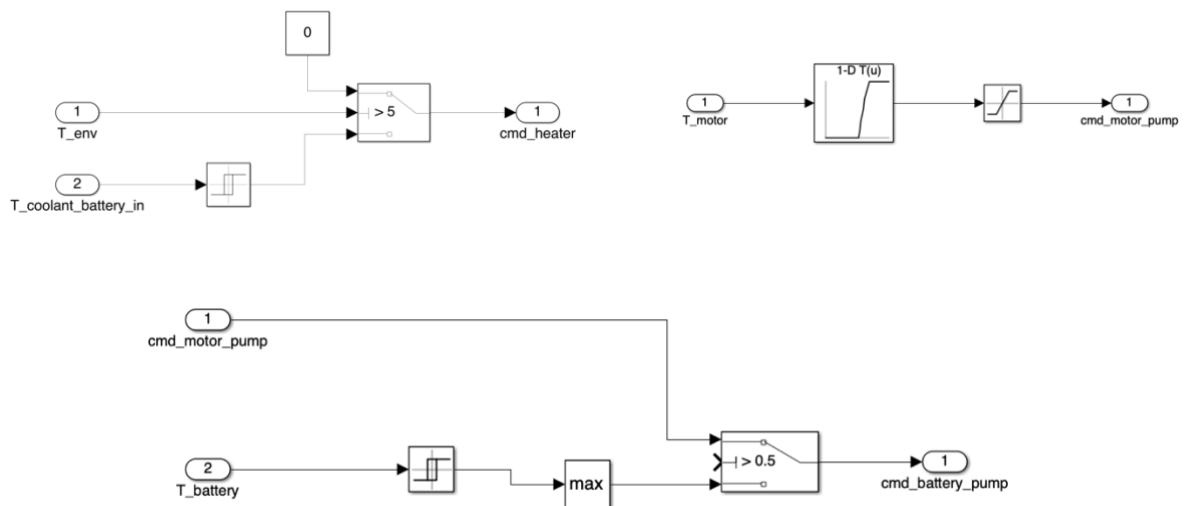


Figure 19: Coolant loop control

The control logic for the coolant loop is essential as it plays a crucial role in maintaining optimal operating conditions for the powertrain and energy storage systems. This intricate system encompasses the battery pump, motor pump, and battery heater and is designed to ensure efficient thermal regulation.

In Figure 19, the control block is dedicated to managing the battery heater. It does so by comparing the environmental temperature (T_{env}) and the coolant inlet temperature for the battery ($T_{coolant_battery_in}$). When the environmental temperature falls below a certain threshold (0°C), and the temperature difference between the coolant and the environment exceeds (5°C), the system activates the battery heater (cmd_heater). This proactive mechanism ensures that the battery is preheated in cold conditions, thus mitigating efficiency losses attributed to low temperatures. The focus shifts to motor pump control. The motor's temperature (T_{motor}) is input into a transfer function, likely representing a temperature-dependent threshold. Whenever the motor temperature surpasses this threshold, the output triggers the motor pump (cmd_motor_pump) to circulate coolant. This action is essential in maintaining the motor's temperature within an optimal range, thereby preventing overheating and ensuring consistent performance.

In Figure 19, the battery temperature ($T_{battery}$) is carefully monitored and processed through a filter, and its output is compared to a threshold. The command for the battery pump ($cmd_battery_pump$) is then determined by the higher of the two inputs—the motor pump command or the battery temperature condition. In essence, if either the motor pump is active or the battery temperature exceeds the threshold, the battery pump is activated to circulate coolant. This integrated control strategy is critical in ensuring the battery remains within a safe operating temperature by dynamically adjusting the coolant flow based on real-time thermal demands. The overall requirement is to achieve a certain temperature and to maintain other component temperatures within their threshold.

4 Chapter 4:

Results and Discussions

4.1 Cabin Heating

The graph illustrates how an electric vehicle's heating system warms up the cabin in cold weather. The x-axis represents time in seconds, and the y-axis shows temperature in degrees Celsius. Three lines are shown: the cabin temperature (red line), the set cabin temperature (blue line), and the outside temperature (yellow line). Initially, the cabin temperature matches the outside temperature of -10°C . The heating system quickly warms the cabin, and as it approaches the setpoint of 20°C , the rate of warming slows down to avoid overheating. Once it reaches 20°C , the cabin temperature fluctuates slightly around that point, showing that the heating system is working to maintain a steady temperature despite the cold outside. Overall, the graph demonstrates that the heating system effectively warms the cabin in cold weather and maintains the desired temperature, indicating the reliability of the control strategies

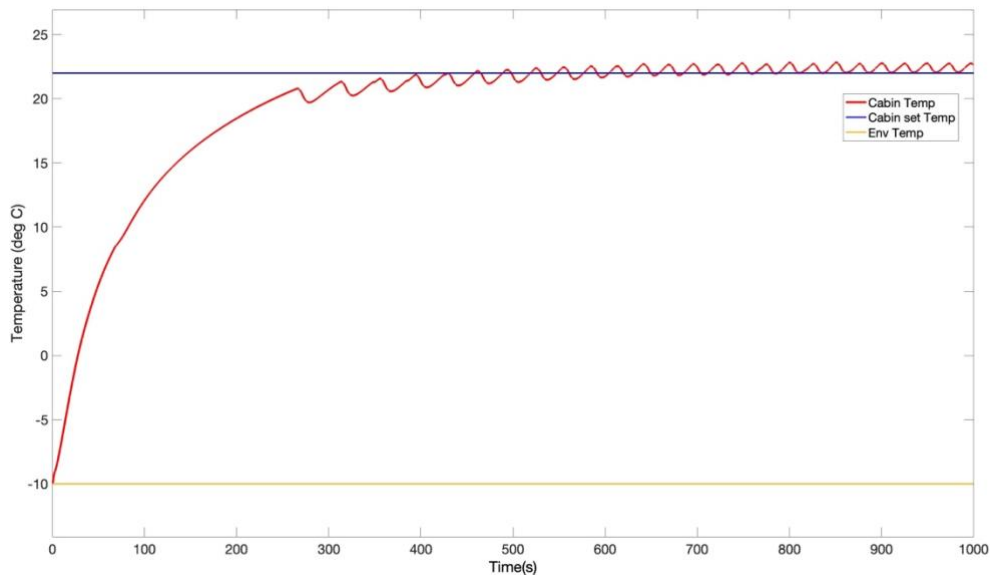


Figure 20: Cabin temperature

4.2 Battery Temperature

A battery pump pumps the coolant into the battery to achieve the desired battery temperature. It takes 1000s for the battery to reach its targeted temperature value of 20 degrees C. Figure 21 depicts the temperature changes in a battery as it is heated from -10°C to 20°C. Initially, the temperature rises steadily as the heating system works to bring the battery to an efficient operational temperature. As the battery approaches the target temperature, the rate of increase slows down, indicating the system's modulation to prevent overshoot. Eventually, the battery temperature closely aligns with the target temperature, demonstrating the effectiveness of the thermal management system in achieving and maintaining the desired operational temperature, even under harsh environmental conditions.

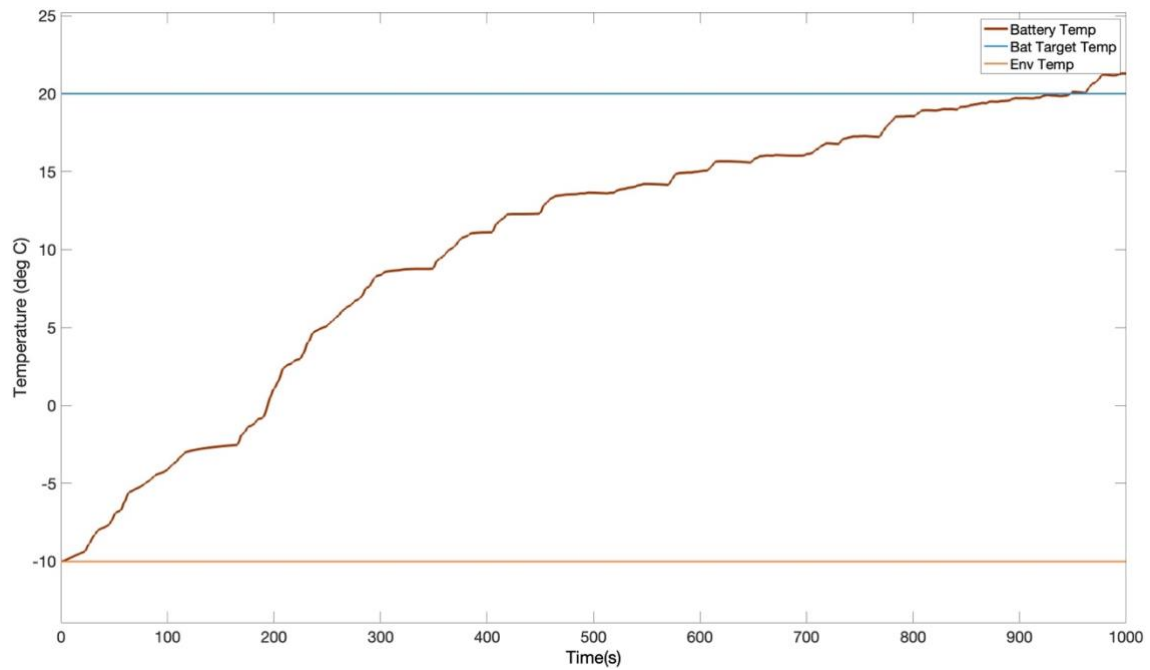


Figure 21: Battery target temperature

Battery heat generation rate is calculated based on the current I_{bat} and internal resistance R_{int} as follows

$$q_{bat} = I_{bat}^2 * R_{int} [3]$$

4.3 Heat Flow Rate

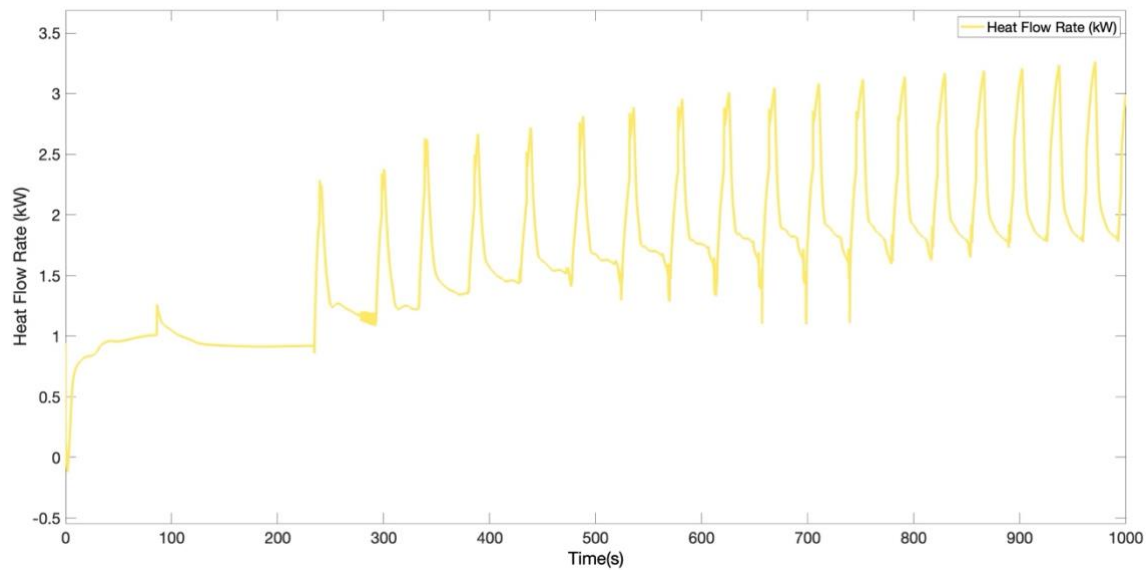


Figure 22: Overall Heat Flow Rate

The heat flow rate of the thermal management system's components, condenser, radiator and fan, cabin, evaporator, and chillers released in a -10°C environment while providing heat. The system keeps the vehicle's interior and critical parts, like the battery and motor, at the right temperature in the

The heat flow rate sharply increases initially, showing the system reacts fast to the cold start. This means much heat is made and moved quickly to warm up the cabin and battery in low temperatures. After the initial rise, the heat release becomes steady, maintaining a consistent warmth of about 0.8 to 1.0 kW. This stable period shows the system can sustain a steady warmth to fight the extreme cold and keep the cabin comfy and the battery working right. The up-and-down movement is seen after the heat stabilises and the system adjusts to changing heating needs. These changes might be due to the cycling of heating parts and the controlled work of the radiator and fan to stop overheating and save energy. In general, these results show the thermal management system effectively handles heat in cold conditions. The system keeps the correct heat levels to warm the vehicle's inside and parts, even at -10°C . The up and down movements show the system adjusts well to keep steady and efficient warmth.

4.3.1 Individual component heat flow rate

The Figure 23 shows the heat flow rate of the components, such as the condenser, evaporator, chiller, cabin HX, and radiator. At ambient -10°C , the evaporator has the highest heat flow rate due to its function in heating the cabin. The value rises to 2.5 kW, the highest among the other components. The fluctuations in the evaporator heat flow rate demonstrate how the system adjusts to the vehicle's heating needs in real-time. The purple line represents the Coolant Radiator Heat, which remains steady at around 0.6 kW, suggesting consistent heat dissipation to the environment. This is likely due to the coolant loop, which is crucial for maintaining thermal equilibrium in the system. The Cabin Heat Exchange to the Exterior shows minimal fluctuations around zero, indicating that heat losses to the exterior are controlled and minor compared to the active heating processes within the cabin.

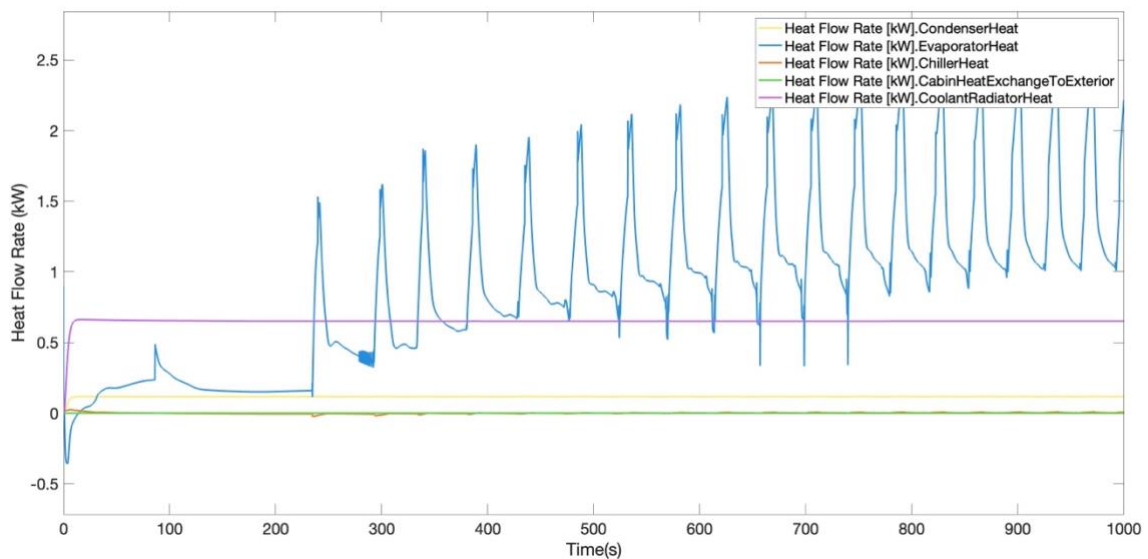


Figure 23: Individual Heat flow rate

4.4 Power Consumption

Figure 24. shows the overall and individual power consumed by the refrigeration components of the system. When the heating system starts up in cold conditions, there is a noticeable increase in power usage due to the compressor's operation, peaking at around 6 kW. This reflects the significant energy needed to kickstart

the thermal management processes. Once the system stabilises, power consumption fluctuates in a predictable pattern, showing how components like the compressor and blower fan adjust their operations to maintain the required temperatures inside the vehicle's cabin and battery.

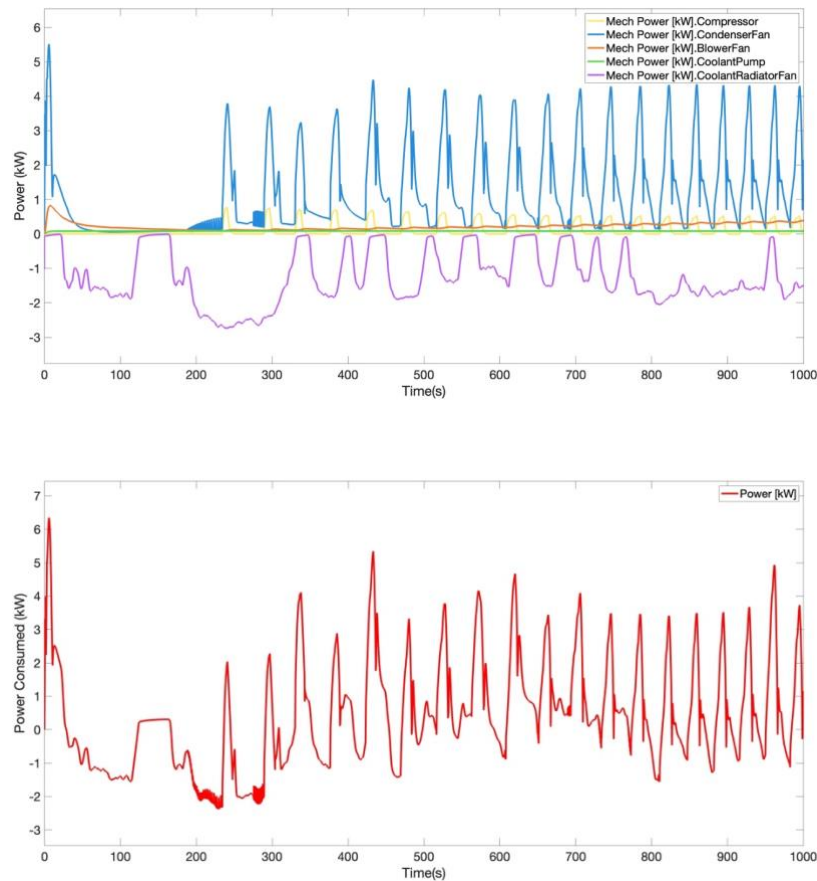


Figure 24: Individual and overall power consumption

Figure 24 indicates an underlying issue with the compressor cycling, which could be a sign of inefficiencies in system control or unnecessary power consumption if not managed properly. Furthermore, the negative power values in the coolant radiator fan may indicate periods of regeneration or errors in power calculation that need to be investigated. Overall, while the system is operating, there are opportunities to optimise the control strategy to reduce power spikes and enhance energy efficiency. This can be achieved by minimising compressor

cycling and ensuring that all components contribute effectively to the system's thermal management objectives.

4.5 Future work in the EVTMS model

4.5.1 EVTMS Control

The EVTMS model is designed to comprehend the thermodynamic process within the refrigeration cycle of a thermal management system. Its purpose is to ensure cabin comfort for the HVAC system. The cabin heating process relies on a PTC heater, and its control logic involves employing a bang-bang control logic, PID controller, and relay switches to monitor component temperatures. This setup allows for better control of the PTC heater and the current demand into the battery.

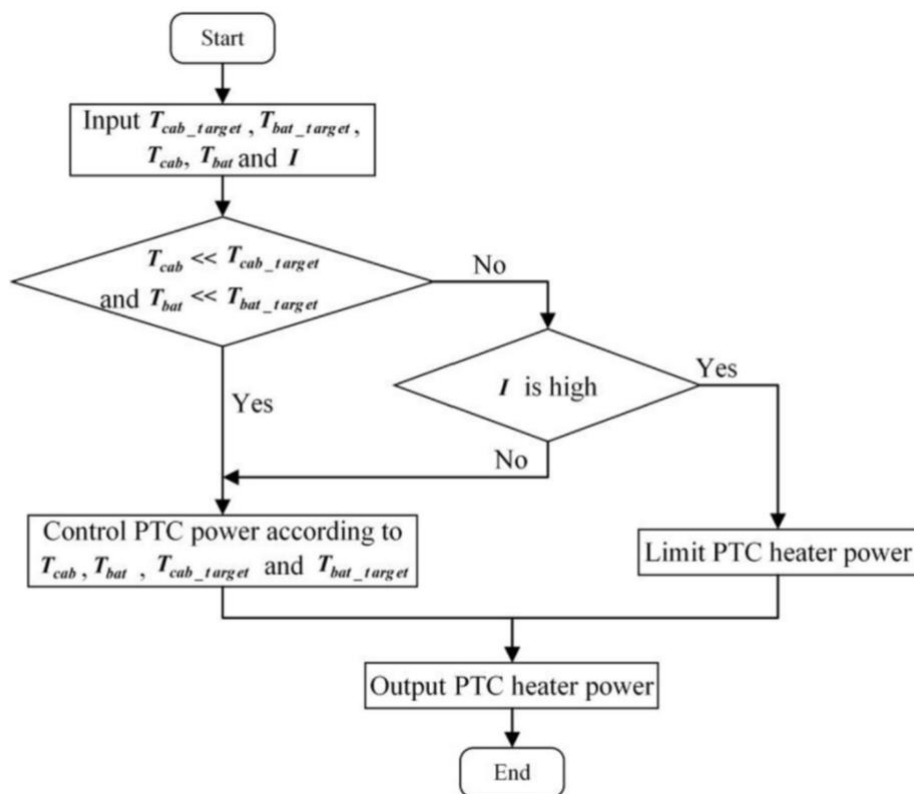


Figure 25: Control system flow chart

Error! Reference source not found. shows the control flow diagram for improved system control logic. This control scheme limits the PTC power consumption to achieve cabin comfort and allows the battery to meet its target temperature[26], [27], [28], [29].

4.5.2 Energy and Exergy Modelling

The energy modelling will be improved to capture all energy interactions within the TMS fully. The current model includes components like the compressor, condenser, evaporator, coolant pump, and various fans, but it will now be expanded to consider all forms of energy exchange, including traditionally overlooked factors like parasitic losses, thermal resistances, and energy recycling within the system. This detailed energy modelling will provide an accurate assessment of the total energy input versus useful energy output, helping to identify areas with the most significant energy losses. The model will also investigate the effects of different environmental conditions, such as extreme temperatures, on the system's energy performance, ensuring that the TMS can maintain optimal vehicle operation in various scenarios[30], [31].

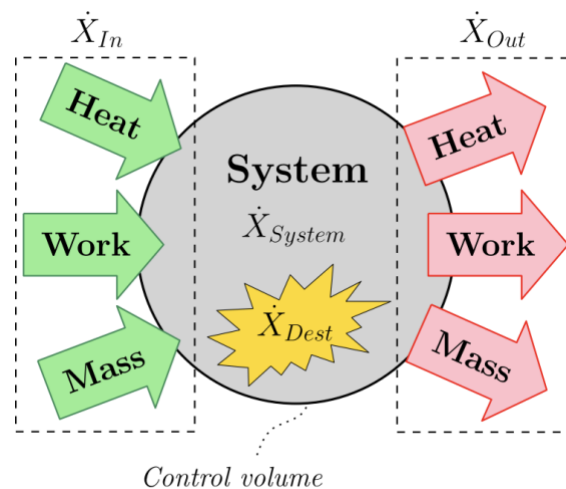


Figure 26: Exergy modelling

The TMS framework will incorporate exergy modelling alongside energy analysis. Exergy, which assesses the quality of energy and its ability to perform work, offers valuable insights into the true efficiency of the TMS. The exergy model will identify irreversible processes within the system, such as heat transfer and fluid friction, leading to entropy generation. By quantifying exergy destruction in each component, the model will pinpoint significant sources of inefficiency, such as the compressor and condenser. This analysis will drive modifications to the TMS design, including the integration of more efficient heat exchangers, advanced control strategies, and the optimisation of coolant flow rates to reduce exergy losses[30], [31]. In addition, the exergy model will consider the overall system's exergy efficiency, providing insights into how different TMS components interact and contribute to vehicle thermal management. This comprehensive approach will evaluate exergy efficiency under various operating conditions, ensuring high efficiency across heating, cooling, and battery thermal management modes. Furthermore, the integration of exergy analysis will support the development of strategies for waste heat recovery, enhancing the vehicle's overall energy efficiency by converting lost energy into useful work.

- Exergy Balance:

$$X_{sys} = X_{in} - X_{out} + X_{dest}$$

4.5.3 Waste Heat Recovery Methods

Research on waste heat recovery(WHR) for electric and hybrid vehicles underscores the importance of integrating advanced thermal management techniques to enhance energy efficiency and reduce emissions. The focus is on leveraging waste heat from components such as motors and batteries, especially in cold climates where traditional heating methods are less effective. One proposed approach involves a three-heat source segmented heating control strategy aimed at optimising the use of waste heat for cabin and battery heating, resulting in significant energy savings[18]. Another area of interest is the exploration of technologies like thermoelectric generators and organic Rankine cycles to capture and utilise waste heat, thereby improving overall powertrain

efficiency[32], [33]. While the integration of these systems holds the potential for substantial energy savings and enhanced vehicle performance, it is important to note challenges such as increased complexity and cost. The findings suggest that future advancements in vehicle thermal management should prioritise maximising waste heat utilisation to achieve superior energy efficiency and environmental performance[18][34].

5 Chapter 5:

Conclusion

The energy consumption of electric vehicles in low-temperature environments has recently attracted significant attention. The thermal management system of an EV plays a vital role in energy consumption from various heat-generating sources. This paper discusses the design approach to the thermal management system, which consists of a refrigeration loop, a coolant loop and an HVAC loop. The simulation work is done to study the thermal behaviour of the system at low ambient temperatures. The analysis of the thermal management system (TMS) illustrates its ability to effectively regulate the target temperature with minimal oscillations, showcasing a highly efficient control strategy that can swiftly adapt to fluctuations in heat flow rates across the system's components. This stability is a critical indication of the TMS's capacity to ensure consistent thermal comfort in challenging conditions, which is vital for maintaining passenger comfort and safeguarding temperature-sensitive components such as the battery.

Regarding energy consumption, the power consumption graphs demonstrate the dynamic nature of the TMS, with spikes corresponding to the activation of various heating components such as the compressor, condenser fan, and coolant pump. The system also exhibits periods of negative power consumption, likely attributable to regenerative effects or energy recovery strategies integrated within the TMS. However, the presence of high-power consumption peaks, especially during the initial heating phase, highlights potential areas for optimisation. These peaks suggest that the system could benefit from advanced control strategies or component design improvements to minimise these surges, thereby enhancing overall energy efficiency. The steady-state power consumption after the initial surge indicates that the system is capable of maintaining the desired thermal conditions with relatively lower energy input, representing a positive outcome for prolonged operation scenarios.

The heat flow rate analysis further bolsters these findings by revealing the distinct contributions of different components to the overall thermal balance of the system. The condenser and coolant radiator play significant roles in dissipating excess heat to prevent overheating. The chiller, primarily designed for cooling, exhibits a consistent, albeit low, contribution to heat flow, indicating its involvement in maintaining system equilibrium. The interaction between these components suggests a well-coordinated TMS that efficiently manages heating and cooling demands. The results also emphasise the importance of optimising each component's operation to ensure that the system can flexibly respond to varying thermal loads without compromising efficiency. Future endeavours should concentrate on refining the energy and exergy models to reduce energy consumption peaks and enhance the system's capacity to recover and utilise waste heat effectively. This approach will not only improve the EVTMS's performance but also contribute to the sustainability of the entire vehicle system by reducing its overall energy footprint.

References

- [1] H. Zou, G. Huang, S. Shao, X. Zhang, C. Tian, and X. Zhang, 'Experimental Study on Heating Performance of an R1234yf Heat Pump System for Electric Cars', in *Energy Procedia*, Elsevier Ltd, 2017, pp. 1015–1021. doi: 10.1016/j.egypro.2017.12.348.
- [2] I. Energy Agency, 'Global EV Outlook 2024 Moving towards increased affordability', 2024. [Online]. Available: www.iea.org
- [3] B. Xu and Z. Arjmandzadeh, 'Parametric study on thermal management system for the range of full (Tesla Model S)/ compact-size (Tesla Model 3) electric vehicles', *Energy Convers Manag*, vol. 278, Feb. 2023, doi: 10.1016/j.enconman.2023.116753.
- [4] A. Yokoyama, T. Osaka, Y. Imanishi, and S. Sekiya, 'Thermal Management System for Electric Vehicles', *SAE International Journal of Materials and Manufacturing*, vol. 4, no. 1, pp. 1277–1285, Apr. 2011, doi: 10.4271/2011-01-1336.
- [5] D. Dan, Y. Zhao, M. Wei, and X. Wang, 'Review of Thermal Management Technology for Electric Vehicles', Jun. 01, 2023, *MDPI*. doi: 10.3390/en16124693.
- [6] J. Hong, J. Song, U. Han, H. Kim, H. Choi, and H. Lee, 'Performance investigation of electric vehicle thermal management system with thermal energy storage and waste heat recovery systems', *eTransportation*, vol. 20, May 2024, doi: 10.1016/j.etrans.2024.100317.
- [7] T. Rana and Y. Yamamoto, 'Universal electric vehicle thermal management system', in *SAE Technical Papers*, SAE International, Aug. 2020. doi: 10.4271/2020-28-0002.
- [8] Y. H. Hung, Y. F. Lue, and H. J. Gu, 'Development of a Thermal Management System for Energy Sources of an Electric Vehicle',

- IEEE/ASME Transactions on Mechatronics*, vol. 21, no. 1, pp. 402–411, Feb. 2016, doi: 10.1109/TMECH.2015.2454851.
- [9] J. Jaguemont, L. Boulon, Y. Dube, and F. Martel, 'Thermal Management of a Hybrid Electric Vehicle in Cold Weather', *IEEE Transactions on Energy Conversion*, vol. 31, no. 3, pp. 1110–1120, Sep. 2016, doi: 10.1109/TEC.2016.2553700.
 - [10] H. Min, Z. Zhang, W. Sun, Z. Min, Y. Yu, and B. Wang, 'A thermal management system control strategy for electric vehicles under low-temperature driving conditions considering battery lifetime', *Appl Therm Eng*, vol. 181, Nov. 2020, doi: 10.1016/j.applthermaleng.2020.115944.
 - [11] M. S. H. Lipu *et al.*, 'A review of state of health and remaining useful life estimation methods for lithium-ion battery in electric vehicles: Challenges and recommendations', Dec. 20, 2018, *Elsevier Ltd.* doi: 10.1016/j.jclepro.2018.09.065.
 - [12] 'Thermal System for Electric Vehicles with Coolant-based Heat Pump'.
 - [13] M. Sarmiento-Carnevali, A. Fly, and P. Piecha, 'Electric Vehicle Cold Start Range Estimation through Battery-in-Loop Simulations within a Virtual Driving Environment', in *SAE Technical Papers*, SAE International, Apr. 2020. doi: 10.4271/2020-01-0453.
 - [14] L. Nicoletti, 'Design and Analyze a Battery Electric Vehicle with Thermal Management'.
 - [15] H. S. Hamut, I. Dincer, and G. F. Naterer, 'Performance assessment of thermal management systems for electric and hybrid electric vehicles', *Int J Energy Res*, vol. 37, no. 1, pp. 1–12, Jan. 2013, doi: 10.1002/er.1951.
 - [16] D. Marcos, F. J. Pino, C. Bordons, and J. J. Guerra, 'The development and validation of a thermal model for the cabin of a vehicle', *Appl Therm Eng*, vol. 66, no. 1–2, pp. 646–656, 2014, doi: 10.1016/j.applthermaleng.2014.02.054.

- [17] A. Warey, S. Kaushik, B. Khalighi, M. Cruse, and G. Venkatesan, 'Data-driven prediction of vehicle cabin thermal comfort: using machine learning and high-fidelity simulation results', *Int J Heat Mass Transf*, vol. 148, Feb. 2020, doi: 10.1016/j.ijheatmasstransfer.2019.119083.
- [18] X. A. Liu, F. Zhang, Z. Zhang, Y. Huang, L. Chen, and X. Li, 'A three-heat source segmented heating control strategy based on waste heat recovery technology for electric vehicles', *Energy Convers Manag*, vol. 300, Jan. 2024, doi: 10.1016/j.enconman.2023.117932.
- [19] W. Li *et al.*, 'Performance evaluation of secondary loop low-temperature heat pump system for frost prevention in electric vehicles', *Appl Therm Eng*, vol. 182, Jan. 2021, doi: 10.1016/j.applthermaleng.2020.115615.
- [20] T. Kiss, J. Lustbader, and D. Leighton, 'Modeling of an Electric Vehicle Thermal Management System in MATLAB/Simulink', in *SAE Technical Papers*, SAE International, Apr. 2015. doi: 10.4271/2015-01-1708.
- [21] K. Li *et al.*, 'Investigation on the performance and characteristics of a heat pump system for electric vehicles under extreme temperature conditions', *Case Studies in Thermal Engineering*, vol. 27, Oct. 2021, doi: 10.1016/j.csite.2021.101308.
- [22] K. Li, J. Yu, R. Yu, L. Su, Y. Fang, and Z. Yang, 'Experimental investigation on heating performance of newly designed air source heat pump system for electric vehicles', *J Therm Sci Eng Appl*, vol. 13, no. 2, Apr. 2021, doi: 10.1115/1.4047821.
- [23] 'GitHub - Florin-Nae/design-and-test-a-thermal-management-system-for-a-vehicle: Design and Test a Thermal Management System for a Vehicle'.
- [24] G. Previati, G. Mastinu, and M. Gobbi, 'Thermal Management of Electrified Vehicles—A Review', Feb. 01, 2022, *MDPI*. doi: 10.3390/en15041326.
- [25] 'Electric Vehicle Thermal Management- MATLAB & Simulink- MathWorks United Kingdom'.

- [26] D. Dvorak, D. Basciotti, and I. Gellai, 'Demand-based control design for efficient heat pump operation of electric vehicles', *Energies (Basel)*, vol. 13, no. 20, Oct. 2020, doi: 10.3390/en13205440.
- [27] Q. Xu *et al.*, 'Experimental investigation on control strategy of electric vehicle heat pump with refrigerant injection at low temperature', *International Journal of Refrigeration*, vol. 138, pp. 40–51, Jun. 2022, doi: 10.1016/j.ijrefrig.2022.03.024.
- [28] C. L. Weng and L. J. Kau, 'Design and implementation of a low-energy-consumption air-conditioning control system for smart vehicle', *J Healthc Eng*, vol. 2019, 2019, doi: 10.1155/2019/3858560.
- [29] Y. Xie *et al.*, 'An intelligent eco-heating control strategy for heat-pump air conditioning system of electric vehicles', *Appl Therm Eng*, vol. 216, Nov. 2022, doi: 10.1016/j.applthermaleng.2022.119126.
- [30] H. S. Hamut, I. Dincer, and G. F. Naterer, 'Exergy analysis of a TMS (thermal management system) for range-extended EVs (electric vehicles)', *Energy*, vol. 46, no. 1, pp. 117–125, 2012, doi: 10.1016/j.energy.2011.12.041.
- [31] F. Dettù, G. Pozzato, D. M. Rizzo, and S. Onori, 'Exergy-based modeling framework for hybrid and electric ground vehicles', *Appl Energy*, vol. 300, Oct. 2021, doi: 10.1016/j.apenergy.2021.117320.
- [32] S. Miller, P. Manager, M. Dr-Ing Lorenzo Nicoletti, and S. Application Engineer, 'A Holistic Approach for Designing a Battery Electric Vehicle Thermal Management System With a Focus on Electrothermal Modelling of the Traction Battery', 2023.
- [33] Z. Tian, B. Gu, W. Gao, and Y. Zhang, 'Performance evaluation of an electric vehicle thermal management system with waste heat recovery', *Appl Therm Eng*, vol. 169, Mar. 2020, doi: 10.1016/j.applthermaleng.2020.114976.

- [34] S. Singh, M. Jennings, S. Katragadda, J. Che, and N. Miljkovic, 'System design and analysis methods for optimal electric vehicle thermal management', *Appl Therm Eng*, vol. 232, Sep. 2023, doi: 10.1016/j.applthermaleng.2023.120990.

Appendices

Appendix A Ethical approval letter



8 May 2024

Dear Mr Sureshkumar ,

Reference: CURES/21937/2024

Project ID: 25288

Title: Efficient Thermal/Energy management in EVs at low temperature

Thank you for your application to the Cranfield University Research Ethics System (CURES).

We are pleased to inform you your CURES application, reference CURES/21937/2024 has been reviewed. You may now proceed with the research activities you have sought approval for.

If you have any queries, please contact CURES Support.

We wish you every success with your project.

Regards,

CURES Team

Appendix B

CoolingPower_kW = 3;

Cp = 1005;

Cp_air = 1.005;

P = 101325;

R = 287.05;

RH_cabin_init = 0.5;

RH_env = 0.5;

RH_final = 50;

RH_initial = 50;

RPM_Compressor_nominal = 1000;

RPM_CondFan = 3000;

RPM_EvapFan = 3000;

T_bat_target = 20;

T_cabin_init = -10;

T_charger_init = -10;

T_dcdc_init = -10;

T_env = -10;

T_final = 20;

T_initial = -10;

T_inverter_init = -10;

T_motor_init = -10;

T_set = 20;

Time = 60;

V = 4;

VolFlowRate_CoolantRadiatorFan = 0.5;

VolFlowRate_EvapFan = 0.48735912275357907;

VolFlowrate_CondenserFan = 0.4081632653061224;

air_density = 1.225;

battery_C1_LUT = ...

[1913.6 12447 30609;

4625.7 18872 32995;

23306 40764 47535;

10736 18721 26325;

18036 33630 48274;

12251 18360 26839;

9022.9 23394 30606];

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[3.4966 3.5057 3.5148;  
 3.5519 3.566 3.5653;  
 3.6183 3.6337 3.6402;  
 3.7066 3.7127 3.7213;  
 3.9131 3.9259 3.9376;  
 4.0748 4.0777 4.0821;  
 4.1923 4.1928 4.193];
```

```
battery_N_cells = 20;
```

```
battery_Qe_init = 0;
```

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battery_R0_LUT = ...  
[0.0117 0.0085 0.009;  
 0.011 0.0085 0.009;  
 0.0114 0.0087 0.0092;  
 0.0107 0.0082 0.0088;  
 0.0107 0.0083 0.0091;  
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 0.0116 0.0085 0.0089];
```

```
battery_R1_LUT = ...  
[0.0109 0.0029 0.0013;  
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 0.0034 0.0016 0.001;  
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 0.0033 0.0018 0.0011;  
 0.0028 0.0017 0.0011];
```

```
battery_SOC_LUT = [0; 0.1; 0.25; 0.5; 0.75; 0.9; 1];
```

```
battery_T_init = -10;
```

battery_capacity_LUT = [280.081 276.25 276.392];

battery_cell_cp = 795;

battery_cell_mass = 2.5;

battery_temperature_LUT = [-10 0 5 20 40];

cabin_CO2_init = 0.0004;

cabin_H = 1.5;

cabin_L = 2;

cabin_RH_init = 0.5;

cabin_T_init = -10;

cabin_W = 2;

cabin_doors_conductivity = 0.2;

cabin_duct_area = 0.04;

cabin_glass_conductivity = 1;

cabin_p_init = 0.101325;

cabin_roof_conductivity = 0.2;

condenser_air_massflowrate = 0.5;

condenser_inlet_enthalpy = 457;

condenser_inlet_p = 2.734;

condenser_inlet_temp = 40;

condenser_subcooling_temp = 5;

coolant_T_init = 30;

coolant_channel_D = 0.0092;

coolant_circuit_length = 5;

coolant_circuit_thermal_mass = 20;

coolant_density = 1092;

coolant_p_init = 0.101325;

coolant_pipe_D = 0.02;

coolant_tank_area = 0.0121;

coolant_tank_volume = 5;

delta_h = 30301.028937527793;

door_or_window_area = 0.5;

duct_W = 0.2;

e_sat_final = 23.369471234064431;

e_sat_initial = 2.8676958564508803;

evap_air_massflowrate = 0.59701492537313439;

evap_inlet_enthalpy = 267;

evap_inlet_p = 0.934;

evap_outlet_enthalpy = 430;

evap_superheat_temp = 5;

evap_temperature_drop = 5;

h_fg_final = 2.41732E+6;

h_fg_initial = 2.54284E+6;

h_final = 294789.16110008542;

h_initial = 264488.13216255762;

heat_load = 162582.22185907542;

heat_load_kJ = 162.58222185907542;

heating_power = 2709.7036976512568;

heating_power_kW = 2.709703697651257;

m_air = 5.3655676905980414;

moist_air_RH_init = 0.5;

moist_air_T_init = -10;

```
moist_air_p_init = 0.101325;

nominal_condens_temp = 45;

nominal_evap_temp = 5;

omega_final = 7.1736923570490382E-5;

omega_initial = 8.8020333790629022E-6;

out = saveVarsMat.out;

pump_displacement = 0.02;

pump_nominal_massflowrate = 1.092;

pump_speed_max = 3000;

radiator_H = 0.2;

radiator_L = 0.6;

radiator_N_fins = 7199.9999999999991;

radiator_N_tubes = 25;

radiator_W = 0.015;

radiator_air_area_fins = 1.4625;

radiator_air_area_flow = 0.097499999999999989;

radiator_air_area_primary = 0.43687499999999996;
```

radiator_air_mass_flow_rate = 0.6125;

radiator_coolant_mass_flow_rate = 1;

radiator_cooling_power = 3;

radiator_fin_spacing = 0.002;

radiator_gap_H = 0.0067708333333333336;

radiator_tube_H = 0.0015;

radiator_tube_Leq = 1.9;

radiator_wall_conductivity = 240;

radiator_wall_thickness = 0.0001;

refrigerant_T_init = -10;

refrigerant_alpha_init = 0.76;

refrigerant_massflowrate = 0.018404907975460124;

refrigerant_p_init = 1;

rho_air_initial = 1.3413919226495103;

tube_D = 0.01;

Load calculations:

% Given data

```

T_initial = -10; % Initial temperature in Celsius
RH_initial = 50; % Initial relative humidity in percentage
T_final = 20; % Final temperature in Celsius
RH_final = 50; % Final relative humidity in percentage
P = 101325; % Pressure in Pa
V = 4; % Volume of humid air in cubic meters
Time = 60; % Time duration in seconds

% Constants and properties
R = 287.05; % Specific gas constant for dry air in J/(kg·K)
Cp = 1005; % Specific heat capacity of dry air at constant pressure in J/(kg·K)

% Calculate saturation vapor pressure using Magnus equation
e_sat_initial = 6.112 * exp((17.67 * T_initial) / (T_initial + 243.5)); % in Pa
e_sat_final = 6.112 * exp((17.67 * T_final) / (T_final + 243.5)); % in Pa

% Calculate specific humidity at initial and final conditions
omega_initial = 0.622 * (RH_initial / 100) * e_sat_initial / (P - (1 - (RH_initial / 100)) * e_sat_initial); % in kg/kg dry air
omega_final = 0.622 * (RH_final / 100) * e_sat_final / (P - (1 - (RH_final / 100)) * e_sat_final); % in kg/kg dry air

% Calculate latent heat of vaporization at initial and final temperatures (converted to J/kg)
h_fg_initial = 2501000 - 4184 * T_initial; % Latent heat of vaporization at initial temperature in J/kg
h_fg_final = 2501000 - 4184 * T_final; % Latent heat of vaporization at final temperature in J/kg

% Calculate enthalpies at initial and final conditions
h_initial = Cp * (T_initial + 273.15) + omega_initial * h_fg_initial; % Initial enthalpy in J/kg dry air

```

```
h_final = Cp * (T_final + 273.15) + omega_final * h_fg_final; % Final enthalpy in  
J/kg dry air
```

```
% Calculate change in enthalpy
```

```
delta_h = h_final - h_initial; % Change in enthalpy in J/kg dry air
```

```
% Mass flow rate of air assuming standard conditions
```

```
rho_air_initial = P / (R * (T_initial + 273.15)); % Density of air at initial condition in  
kg/m^3
```

```
m_air = rho_air_initial * V; % Mass of air in kg
```

```
% Total heat load
```

```
heat_load = delta_h * m_air; % Total heat load in J
```

```
% Convert heat load to kJ
```

```
heat_load_kJ = heat_load / 1000; % Total heat load in kJ
```

```
% Heating power
```

```
heating_power = heat_load / Time; % Heating power in W
```

```
% Convert heating power to kW
```

```
heating_power_kW = heating_power / 1000; % Heating power in kW
```

```
% Display the results
```

```
fprintf('Total heat load: %.2f kJ\n', heat_load_kJ);
```

```
fprintf('Heating Power: %.2f kW\n', heating_power_kW);
```

Coolant Load calculations:

```
% Given data
```

```
T_initial = -10; % Initial temperature of the coolant in Celsius
```

```
T_final = 20; % Final temperature of the coolant in Celsius
```

```
P = 101325; % Pressure in Pa
```

```
V = 5; % Total volume of coolant mixture in liters
```

```
Time = 1000; % Time duration in seconds
```

% Constants and properties

Cp_EG = 2420; % Specific heat capacity of ethylene glycol in J/(kg·K)

Cp_water = 4186; % Specific heat capacity of water in J/(kg·K)

w_EG = 0.5; % Mass fraction of ethylene glycol

% Convert volume to mass assuming density of water and ethylene glycol mixture

rho_water = 1000; % Density of water in kg/m³

V_total = V / 1000; % Convert volume to cubic meters

m_total = V_total * rho_water; % Total mass of coolant mixture in kg

% Calculate the specific heat capacity of the coolant mixture

Cp_mix = w_EG * Cp_EG + (1 - w_EG) * Cp_water; % Specific heat capacity of the coolant mixture in J/(kg·K)

% Calculate the change in temperature

delta_T = T_final - T_initial; % Change in temperature in Celsius (corrected)

% Calculate the heat load

heat_load = m_total * Cp_mix * delta_T; % Heat load in J

% Convert heat load to kJ

heat_load_kJ = heat_load / 1000; % Total heat load in kJ

% Convert heat load to kW

heating_power_kW = heat_load / (1000 * Time); % Heating power in kW
(renamed from cooling power)

% Display the results

fprintf('Total heat load: %.2f kJ\n', heat_load_kJ);

fprintf('Heating power: %.2f kW\n', heating_power_kW);

